Inheritance in C++

Object-Oriented Programming,
Lecture 7

What is inheritance

- A derived class reuses the members of a base class
- It models an is-a relationship: a Car is a Vehicle
- The derived class can add new members or change inherited behaviour

Syntax

```
class Vehicle {
public:
   int wheels;
};

class Car : public Vehicle {
public:
   int doors;
};
```

Access after public inheritance

Base member	Inside derived class	Outside the class
public	Accessible	Accessible
protected	Accessible	Not accessible
private	Not accessible	Not accessible

Method overriding

- A derived class can redefine a base-class function with the same signature
- Mark the base function virtual so the call is chosen at run time
- Use override in the derived class so the compiler checks the signature

Types of inheritance

- Single: one base, one derived class
- Multilevel: Vehicle, then Car, then SportsCar
- Multiple: one class derives from two bases

Questions?